

VIKULOVO, Russian Federation

WMO: 284810

Lat: 56.817N

Lon: 70.617E

Elev: 78

StdP: 100.39

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.5	(c) -30.8	(d) -36.5	(e) 0.1	(f) -33.3	(g) -33.9	(h) 0.2	(i) -30.7	(j) 7.2	(k) -10.7	(l) 6.4	(m) -10.4	(n) 1.5	(o) 180	(p) 0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.5	(c) 29.5	(d) 19.7	(e) 27.8	(f) 18.9	(g) 26.1	(h) 18.1	(i) 21.2	(j) 27.0	(k) 20.2	(l) 25.6	(m) 19.2	(n) 24.4	(o) 3.0	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.0	24.0	18.2	13.2	22.9	17.2	12.4	22.0	61.7	27.2	58.3	25.6	55.1	24.4	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.1	6.3	5.6	-36.1	32.6	3.6	2.3	-38.7	34.3	-40.8	35.6	-42.8	36.9	-45.5	38.6
			-36.2	22.7	3.6	1.1	-38.8	23.5	-40.9	24.1	-42.9	24.8	-45.6	25.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-17.7	-14.6	-6.0	3.6	11.6	17.1	18.4	15.9	10.1	3.2	-6.5	-13.5		
	DBStd	13.87	7.63	7.30	5.62	5.67	5.52	4.72	3.73	3.97	4.47	5.06	7.08	8.02		
	HDD10.0	3797	858	688	497	203	48	5	0	3	53	216	497	729		
	HDD18.3	6124	1117	921	756	442	217	77	47	97	247	469	747	988		
	CDD10.0	836	0	0	0	11	98	218	261	185	57	5	0	0		
	CDD18.3	122	0	0	0	0	9	41	49	22	2	0	0	0		
	CDH23.3	1149	0	0	0	3	134	409	413	172	18	0	0	0		
	CDH26.7	276	0	0	0	0	30	120	91	33	2	0	0	0		
(14) Wind	WSAvg	2.7	2.5	2.6	3.0	3.1	3.1	2.6	2.3	2.4	2.6	2.8	2.9	2.8		
(15) Precipitation	PrecAvg	463	19	13	19	27	38	60	75	68	51	35	32	27		
	PrecMax	580	40	32	42	53	80	154	194	141	107	87	54	54		
	PrecMin	330	2	1	0	1	6	12	20	26	17	2	8	10		
	PrecStd	61	11	7	11	15	19	36	43	28	21	19	13	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.4	2.2	7.8	22.6	29.6	31.8	30.9	29.9	25.8	18.2	7.1	2.2		
		MCWB	-0.3	0.2	3.5	13.1	16.1	20.1	21.0	20.5	16.1	11.1	5.0	1.0		
	2%	DB	-2.1	-0.1	4.7	18.5	26.7	29.9	29.0	27.2	22.5	15.1	4.8	0.6		
		MCWB	-2.8	-1.3	1.7	10.7	15.3	19.5	20.4	19.4	15.1	9.6	3.5	-0.3		
	5%	DB	-4.2	-2.0	2.9	15.2	24.5	27.8	27.4	25.2	19.6	12.7	3.3	-1.3		
		MCWB	-4.9	-2.9	0.7	9.2	14.2	18.8	19.7	18.4	13.8	8.5	2.2	-2.1		
	10%	DB	-6.4	-4.2	1.5	12.4	21.7	25.5	25.7	23.1	17.5	10.5	1.7	-3.1		
		MCWB	-7.0	-5.1	-0.3	7.5	13.0	17.8	18.9	17.5	12.4	7.4	0.7	-3.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.2	0.4	3.8	14.0	18.1	22.2	22.6	21.6	17.5	12.1	5.4	1.3		
		MCDB	0.3	1.9	7.0	19.7	25.9	29.0	28.9	27.7	23.5	16.5	6.6	2.0		
	2%	WB	-2.7	-1.3	2.1	11.2	16.3	20.7	21.4	20.2	15.7	10.4	3.7	-0.3		
		MCDB	-2.1	-0.2	4.0	17.4	24.7	27.1	27.1	25.3	21.2	14.0	4.7	0.6		
	5%	WB	-4.9	-2.9	1.0	9.6	15.0	19.5	20.5	19.2	14.3	9.0	2.2	-2.0		
		MCDB	-4.2	-1.9	2.7	15.0	23.0	25.7	25.9	24.0	18.9	12.2	3.1	-1.3		
	10%	WB	-7.1	-5.1	-0.2	7.8	13.6	18.4	19.6	18.1	12.9	7.5	0.8	-3.8		
		MCDB	-6.4	-4.3	1.4	12.1	20.6	24.3	24.5	22.1	16.7	10.3	1.5	-3.1		
(35) Mean Daily Temperature Range	MDBR		6.9	8.5	8.7	9.0	11.3	10.2	9.5	9.4	8.8	6.7	5.9	6.9		
	5% DB	MCDBR	7.1	8.1	8.3	13.4	15.1	13.3	12.0	12.5	12.5	10.9	5.2	7.6		
		MCWBR	6.9	7.4	6.1	7.5	6.6	5.5	5.2	5.7	6.3	6.5	4.6	7.1		
	5% WB	MCDBR	6.9	8.0	7.7	12.5	13.8	11.9	10.9	11.3	11.8	9.8	5.2	7.7		
		MCWBR	6.7	7.5	6.0	7.3	6.9	5.5	5.2	5.7	6.3	6.3	4.8	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.277	0.311	0.368	0.413	0.407	0.409	0.425	0.429	0.377	0.353	0.311	0.279		
	taud		2.035	2.019	1.934	2.051	2.229	2.288	2.242	2.198	2.305	2.220	2.087	1.979		
	Ebn at Noon		569	708	756	791	830	833	807	772	762	662	515	437		
	Edh at Noon		61	96	136	143	127	122	125	123	95	79	59	50		
(44) All-Sky Solar Radiation	RadAvg		0.77	1.79	3.31	4.56	5.32	5.68	5.40	4.16	2.76	1.40	0.66	0.47		
	RadStd		0.06	0.12	0.23	0.36	0.41	0.46	0.48	0.38	0.30	0.19	0.11	0.06		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page